# Hydrogen Ground-State Hyperfine Splitting as a Canonical Activation Gap: From the Fermi Contact Interaction to the Schottky Heat-Capacity Anomaly

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**Abstract**

We give a self-contained, orthodox derivation of the hydrogen $1s$ hyperfine splitting starting from the magnetic dipole field (including its distributional contact term) and the resulting Fermi contact Hamiltonian. The leading-order gap $\Delta E_{\rm hfs}$ is expressed in standard constants, recovering the well-known scaling $\Delta E_{\rm hfs} \propto \alpha^4 (m_e/m_p) m_e c^2$. We then treat the $(F=0)$–$(F=1)$ manifold as a two-level system with degeneracies $(g_0, g_1) = (1, 3)$ and derive the exact canonical heat capacity (Schottky anomaly), including its low-temperature asymptotic and an explicit derivation of the peak condition for the degeneracy-asymmetric case. Numerical evaluation yields $\Delta E_{\rm hfs} \approx 5.88\ \mu$eV from the leading term, to be compared with the experimental value 5.8743 $\mu$eV (fractional error $\approx$ 0.097%), with the remaining discrepancy attributable to known QED, recoil, and proton-structure corrections.

## 1 Introduction

The hydrogen ground-state hyperfine splitting provides a concrete, experimentally calibrated example of a gapped two-level subsystem. Independently, the Schottky anomaly describes the characteristic heat-capacity peak arising from discrete-level occupation in the canonical ensemble. Both topics are standard, but they are often presented separately (atomic structure vs. thermal physics). The purpose of this manuscript is to present a single, fully explicit derivation chain that (i) derives the leading-order hyperfine gap from the Fermi contact interaction and (ii) propagates that gap into thermodynamic predictions for the two-level heat capacity, including an explicit derivation of the peak condition for the degeneracy-asymmetric case $(g_0, g_1) = (1, 3)$. Accordingly, the contribution is pedagogical and synthetic rather than one of new fundamental physics: it provides a closed analytic-to-thermodynamic derivation chain from the hydrogen $1s$ hyperfine gap to the exact degeneracy-asymmetric Schottky anomaly, with explicit millikelvin scales relevant to low-temperature spectroscopy and thermodynamics.

## 2 Definitions and standard identities

We use SI units. The fundamental constants are: $c$, $\hbar$, $h = 2\pi\hbar$, $e$, $\varepsilon_0$, $\mu_0 = 1/(\varepsilon_0 c^2)$, $k_B$, and the electron and proton masses $(m_e, m_p)$.

The fine-structure constant is

$$\alpha \equiv \frac{e^2}{4\pi\varepsilon_0 \hbar c}. \tag{1}$$

The Bohr radius is

$$a_0 \equiv \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}. \tag{2}$$

Using (1) (i.e. $e^2 = 4\pi\varepsilon_0\hbar c\,\alpha$), one obtains the standard identity

$$a_0 = \frac{\hbar}{m_e c}\,\frac{1}{\alpha}. \tag{3}$$

The electron and proton magnetic moments are written using $g$-factors:

$$\boldsymbol{\mu}_e = -g_e\mu_B\frac{\mathbf{S}}{\hbar}, \qquad \boldsymbol{\mu}_p = +g_p\mu_N\frac{\mathbf{I}}{\hbar}, \tag{4}$$

where $\mathbf{S}$ is the electron spin operator and $\mathbf{I}$ is the proton spin operator, and

$$\mu_B \equiv \frac{e\hbar}{2m_e}, \qquad \mu_N \equiv \frac{e\hbar}{2m_p}. \tag{5}$$

For hydrogen $1s$, we use $g_e \simeq 2$ and $g_p \simeq 5.5856947$ (proton $g$-factor) for the leading-order estimate; the effect of using the free-electron value $g_e = 2.002319304\ldots$ is discussed in Section 5 [1].

The normalized $1s$ wavefunction is

$$\psi_{1s}(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}}e^{-r/a_0}, \qquad |\psi_{1s}(0)|^2 = \frac{1}{\pi a_0^3}. \tag{6}$$

# 3 Magnetic dipole interaction and the Fermi contact term

## 3.1 Dipole field and contact contribution

The magnetic field of a point dipole $\boldsymbol{\mu}$ at the origin contains a distributional (contact) term. In SI units one may write [2]

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi r^3}\Big[3\hat{\mathbf{r}}(\boldsymbol{\mu}\cdot\hat{\mathbf{r}}) - \boldsymbol{\mu}\Big] + \frac{2\mu_0}{3}\,\boldsymbol{\mu}\,\delta^{(3)}(\mathbf{r}). \tag{7}$$

For an $s$-state, the isotropic contact term is the relevant contribution to the hyperfine splitting.

## 3.2 Fermi contact Hamiltonian and consistent sign convention

To avoid sign ambiguity, we compute the interaction as the energy of the electron magnetic moment in the proton's magnetic field:

$$H_{\text{contact}} = -\boldsymbol{\mu}_e\cdot\mathbf{B}_p^{(\text{contact})}. \tag{8}$$

Using the proton contact field from (7),

$$\mathbf{B}_p^{(\text{contact})} = \frac{2\mu_0}{3}\,\boldsymbol{\mu}_p\,\delta^{(3)}(\mathbf{r}), \tag{9}$$

we obtain

$$H_{\text{contact}} = -\frac{2\mu_0}{3}\,(\boldsymbol{\mu}_e\cdot\boldsymbol{\mu}_p)\,\delta^{(3)}(\mathbf{r}). \tag{10}$$

The electron and proton magnetic moments are

$$\boldsymbol{\mu}_e = -g_e\mu_B\frac{\mathbf{S}}{\hbar}, \qquad \boldsymbol{\mu}_p = +g_p\mu_N\frac{\mathbf{I}}{\hbar}, \tag{11}$$

so that

$$\boldsymbol{\mu}_e\cdot\boldsymbol{\mu}_p = -\,g_e g_p\,\mu_B\mu_N\,\frac{\mathbf{S}\cdot\mathbf{I}}{\hbar^2}. \tag{12}$$

Substituting into (**??**) gives

$$H_{\text{contact}} = +\frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, \frac{\mathbf{S}\cdot\mathbf{I}}{\hbar^2}\, \delta^{(3)}(\mathbf{r}). \tag{13}$$

Taking the expectation value in the $1s$ spatial state yields

$$\langle H_{\text{contact}}\rangle_{1s} = A\,\mathbf{S}\cdot\mathbf{I}, \qquad A \equiv \frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, \frac{|\psi_{1s}(0)|^2}{\hbar^2}, \tag{14}$$

with $A > 0$.

The energy eigenstates are labeled by total spin $\mathbf{F} = \mathbf{S} + \mathbf{I}$ with $S = I = 1/2$ and $F \in \{0, 1\}$. Using

$$\mathbf{S}\cdot\mathbf{I} = \frac{1}{2}\Big(F(F+1) - S(S+1) - I(I+1)\Big)\hbar^2, \tag{15}$$

and $S(S+1) = I(I+1) = 3/4$, one finds

$$\langle \mathbf{S}\cdot\mathbf{I}\rangle_{F=1} = +\frac{1}{4}\hbar^2, \qquad \langle \mathbf{S}\cdot\mathbf{I}\rangle_{F=0} = -\frac{3}{4}\hbar^2. \tag{16}$$

Define

$$\mathcal{A} \equiv A\hbar^2 = \frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, |\psi_{1s}(0)|^2. \tag{17}$$

Then

$$\Delta E_{\text{hfs}} \equiv E_{F=1} - E_{F=0} = \mathcal{A}\left(\frac{1}{4} - \left(-\frac{3}{4}\right)\right) = \mathcal{A}. \tag{18}$$

Since $\mathcal{A} > 0$, the singlet lies below the triplet, and therefore

$$E_{F=0} < E_{F=1}, \qquad \Delta E_{\text{hfs}} > 0.$$

Substituting $|\psi_{1s}(0)|^2 = 1/(\pi a_0^3)$ and the definitions of $\mu_B, \mu_N$ yields the standard leading-order Fermi-contact formula

$$\Delta E_{\text{hfs}} = \frac{2\mu_0}{3}\, g_e g_p\, \mu_B \mu_N\, \frac{1}{\pi a_0^3}. \tag{19}$$

For a closely related pedagogical derivation and sign-consistent discussion, see Griffiths [3].

## 4 Closed-form expression in $\alpha$

We eliminate $\mu_0, \mu_B, \mu_N, a_0$ in favor of $(\alpha, m_e, m_p, c)$.

Use $\mu_0 = 1/(\varepsilon_0 c^2)$ and (5):

$$\mu_0\, \mu_B \mu_N = \frac{1}{\varepsilon_0 c^2}\left(\frac{e\hbar}{2m_e}\right)\left(\frac{e\hbar}{2m_p}\right) = \frac{e^2\hbar^2}{4\varepsilon_0 c^2\, m_e m_p}. \tag{20}$$

Insert into (16):

$$\Delta E_{\text{hfs}} = \frac{g_e g_p}{6\pi}\, \frac{e^2\hbar^2}{\varepsilon_0 c^2\, m_e m_p\, a_0^3}. \tag{21}$$

Now eliminate $a_0$ using (2):

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} \quad\Longrightarrow\quad \frac{1}{a_0^3} = \left(\frac{m_e e^2}{4\pi\varepsilon_0\hbar^2}\right)^3.$$

Substitute:

$$\begin{aligned}\Delta E_{\text{hfs}} &= \frac{g_e g_p}{6\pi}\, \frac{e^2\hbar^2}{\varepsilon_0 c^2\, m_e m_p}\left(\frac{m_e e^2}{4\pi\varepsilon_0\hbar^2}\right)^3 \\ &= \frac{g_e g_p}{6\pi}\, \frac{m_e^2}{m_p}\, \frac{e^8}{(4\pi)^3\, \varepsilon_0^4\, \hbar^4}\, \frac{1}{c^2}.\end{aligned} \tag{22}$$

Table 1: Constants used in the leading-order numerical estimate of $\Delta E_{\rm hfs}^{(0)}$.

| Quantity | Value | Source |
|---|---|---|
| $\alpha$ | $7.2973525643 \times 10^{-3}$ | [1] |
| $m_e c^2$ | 510998.950 eV | [1] |
| $m_e/m_p$ | $5.446170214 \times 10^{-4}$ | [1] |
| $g_e$ | 2 (leading-order) | convention in this paper |
| $g_p$ | 5.5856947 | [1] |

Using $e^2/(4\pi\varepsilon_0) = \alpha\hbar c$, one has

$$\frac{e^8}{(4\pi)^3\varepsilon_0^4} = (4\pi)(\alpha\hbar c)^4,$$

so that

$$\begin{aligned}\Delta E_{\rm hfs} &= \frac{g_e g_p}{6\pi}\,\frac{m_e^2}{m_p}\,\frac{(4\pi)(\alpha\hbar c)^4}{\hbar^4 c^2} \\ &= \frac{4\pi}{6\pi}\,g_e g_p\,\alpha^4\left(\frac{m_e}{m_p}\right)m_e c^2 = \frac{2}{3}\,g_e g_p\,\alpha^4\left(\frac{m_e}{m_p}\right)m_e c^2. \end{aligned} \tag{23}$$

Thus the compact leading-order form is

$$\boxed{\Delta E_{\rm hfs} = \frac{2}{3}\,g_e g_p\,\alpha^4\left(\frac{m_e}{m_p}\right)m_e c^2.} \tag{24}$$

**Literature note (prior derivations).** Equation (20) is the standard leading-order Fermi-contact result ("Fermi energy") and is derived in many classic sources (e.g. Berestetskii–Lifshitz–Pitaevskii) as well as modern reviews. We reproduce the algebra here to keep the manuscript self-contained and to connect the gap to the thermodynamic analysis below [4, 5].

## 5 Numerical evaluation and the 21 cm line

Using CODATA constants and $g_p$ [1], (20) gives a leading-order estimate

$$\Delta E_{\rm hfs}^{(0)} \approx 5.88\ \mu\text{eV}. \tag{25}$$

Experimentally, the hydrogen ground-state hyperfine transition frequency is

$$\nu_{\rm hfs} = 1420.40575177\ \text{MHz}, \tag{26}$$

corresponding to

$$\Delta E_{\rm hfs}^{(\rm exp)} = h\nu_{\rm hfs} \approx 5.8743\ \mu\text{eV}. \tag{27}$$

(See primary measurement references such as Hellwig et al. [6].)

**Fractional deviation and input conventions.** Define

$$\delta \equiv \frac{\Delta E_{\rm hfs}^{(0)} - \Delta E_{\rm hfs}^{(\rm exp)}}{\Delta E_{\rm hfs}^{(\rm exp)}}.$$

Using the leading-order contact formula (20) with the conventional approximation $g_e \simeq 2$ gives $\Delta E_{\rm hfs}^{(0)} \approx 5.88\ \mu$eV and hence

$$\delta \approx 9.7 \times 10^{-4} \approx 0.097\%.$$

If instead one inserts the free-electron value $g_e = 2.002319304\ldots$ into the same leading-order expression, $\Delta E_{\rm hfs}^{(0)}$ shifts by 0.116% and the residual changes at the $\sim 10^{-3}$ level. In precision theory one does not treat (20) as complete: radiative QED corrections, recoil effects, and proton-structure (Zemach-radius) contributions are required for ppm-level agreement [5, 4].

Define the gap temperature scale

$$T_\Delta \equiv \frac{\Delta E_{\rm hfs}^{(\rm exp)}}{k_B} \approx 68.17 \text{ mK}. \tag{28}$$

# 6 Thermodynamics: Schottky anomaly for the hyperfine doublet

## 6.1 Two-level system with degeneracy $(1, 3)$

The $1s$ hyperfine manifold has a singlet ground state ($F = 0$, degeneracy $g_0 = 1$) and a triplet excited state ($F = 1$, degeneracy $g_1 = 3$). Treating the hyperfine degree of freedom as a two-level system with

$$E_0 = 0, \qquad E_1 = \Delta E_{\rm hfs}^{(\rm exp)} \equiv \Delta, \tag{29}$$

the partition function is

$$Z(\beta) = g_0 e^{-\beta E_0} + g_1 e^{-\beta E_1} = 1 + 3e^{-\beta\Delta}, \qquad \beta = \frac{1}{k_B T}. \tag{30}$$

The present treatment assumes that the hyperfine manifold can be regarded as an effectively isolated internal two-level subsystem, which is appropriate for dilute atomic hydrogen under conditions where spin-exchange collisions, molecular formation, and strong external-field splittings do not dominate the equilibration.

## 6.2 Exact heat capacity

The mean energy is

$$\langle E \rangle = \frac{3\Delta e^{-\beta\Delta}}{1 + 3e^{-\beta\Delta}}. \tag{31}$$

Differentiation yields the exact Schottky heat capacity per atom:

$$\boxed{C_V(T) = k_B \left(\frac{\Delta}{k_B T}\right)^2 \frac{3e^{-\Delta/(k_B T)}}{\left(1 + 3e^{-\Delta/(k_B T)}\right)^2}.} \tag{32}$$

For one mole, multiply by Avogadro's number (replace $k_B$ by $R$) [7, 8].

## 6.3 Low-temperature asymptotic (frozen regime)

For $k_B T \ll \Delta$, $e^{-\Delta/(k_B T)} \ll 1$ and (28) reduces to

$$\boxed{C_V(T) \sim 3k_B \left(\frac{\Delta}{k_B T}\right)^2 e^{-\Delta/(k_B T)} \qquad (k_B T \ll \Delta).} \tag{33}$$

In the opposite limit $k_B T \gg \Delta$, the Boltzmann weights of the singlet and triplet levels become nearly equal, the internal entropy approaches $k_B \ln(g_0 + g_1) = k_B \ln 4$, and the heat capacity again tends to zero:

$$C_V(T) \to 0 \qquad (T \to \infty). \tag{34}$$

Thus the Schottky anomaly is localized in an intermediate temperature window around $T \sim T_\Delta$.

### 6.4 Peak location (derived) for general degeneracy ratio

Let $g_0$ be the ground-state degeneracy and $g_1$ the excited-state degeneracy, with gap $\Delta$. Define $r \equiv g_1/g_0$ and $x \equiv \beta\Delta = \Delta/(k_B T)$. Then

$$\frac{C_V}{k_B} = x^2 \frac{re^{-x}}{(1+re^{-x})^2}. \tag{35}$$

Let $y \equiv re^{-x}$ so that $dy/dx = -y$. Writing $(C_V/k_B) = x^2\, y/(1+y)^2$, we obtain

$$\frac{d}{dx}\left(\frac{C_V}{k_B}\right) = 2x\,\frac{y}{(1+y)^2} + x^2\,\frac{d}{dx}\left(\frac{y}{(1+y)^2}\right), \tag{36}$$

$$\frac{d}{dx}\left(\frac{y}{(1+y)^2}\right) = \left(\frac{1}{(1+y)^2} - \frac{2y}{(1+y)^3}\right)\frac{dy}{dx} = \frac{(1-y)}{(1+y)^3}\,\frac{dy}{dx}. \tag{37}$$

Using $dy/dx = -y$ gives

$$\frac{d}{dx}\left(\frac{y}{(1+y)^2}\right) = -\frac{y(1-y)}{(1+y)^3}. \tag{38}$$

Setting the derivative to zero yields the peak condition

$$\boxed{2(1+y) = x(1-y), \qquad y = re^{-x}.} \tag{39}$$

We do not claim originality for the transcendental peak equation itself; it is included here for completeness in the specific hydrogen-hyperfine context. For the physically relevant root one must have $y < 1$, since the right-hand side of (35) is positive only when $1 - y > 0$. For hydrogen hyperfine, $(g_0, g_1) = (1, 3)$ so $r = 3$. The solution of (35) was obtained by standard one-dimensional root-finding (Newton or bisection), giving

$$x_{\text{peak}} \equiv \frac{\Delta}{k_B T_{\text{peak}}} \approx 2.8449, \qquad T_{\text{peak}} \approx 0.3514\, T_\Delta \approx 24.0 \text{ mK}. \tag{40}$$

At this point

$$y_{\text{peak}} = 3e^{-x_{\text{peak}}} \approx 3e^{-2.8449} \approx 0.175 < 1,$$

so the positivity condition is self-consistently satisfied. At the peak one finds $C_V^{\max} \approx 1.02\, k_B$ per atom.

**Interpretation.** The hyperfine degree of freedom behaves thermodynamically as a discrete two-level subsystem: its contribution to $C_V$ is exponentially suppressed for $T \ll T_\Delta$, peaks near $T \sim 0.35\, T_\Delta$, and vanishes again for $T \gg T_\Delta$ as the two levels become nearly equally populated.

## 7 Conclusion

We have presented a single self-contained derivation chain connecting the hydrogen $1s$ hyperfine splitting to the Schottky heat-capacity anomaly of the resulting two-level system. Starting from the dipole field with its contact term, we recovered the standard leading-order Fermi-contact gap and expressed it in closed form in terms of $\alpha$, $m_e$, $m_p$, and the spin $g$-factors. Using the experimentally measured hyperfine energy as the activation gap, we then derived the exact canonical heat capacity for the degeneracy-asymmetric $(g_0, g_1) = (1, 3)$ system, including both low- and high-temperature limits and an explicit peak condition. The resulting millikelvin scale $T_\Delta$ and the peak temperature $T_{\text{peak}}$ provide a direct bridge between atomic spectroscopy and low-temperature thermodynamics. Natural extensions include hydrogen-like ions, Zeeman-resolved hyperfine manifolds, and other discrete nuclear-spin systems.

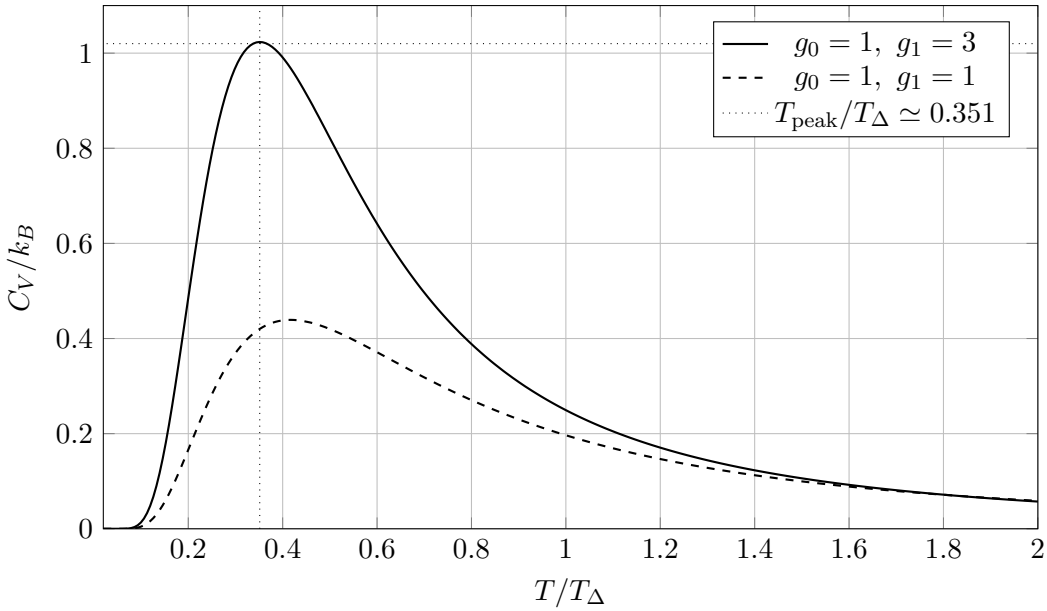


Figure 1: Dimensionless Schottky heat capacity $C_V/k_B$ versus reduced temperature $T/T_\Delta$ for the hydrogen hyperfine doublet $(g_0, g_1) = (1, 3)$, compared with the symmetric two-level case $(g_0, g_1) = (1, 1)$. The asymmetric case peaks at $T_{\text{peak}}/T_\Delta \simeq 0.351$ with $C_V^{\max}/k_B \simeq 1.02$, whereas the symmetric case peaks at the larger temperature ratio $T/T_\Delta \simeq 0.417$.

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